

LAKSHMI VENKATA SAI SARATH

Location: USA | Email: kunapareddyivsaisarathusa@gmail.com | Phone: +1 734 377 2420 | [LinkedIn](#)

SUMMARY

Data Engineer with 8+ years of experience building batch and streaming pipelines using Python, SQL, PySpark, and Apache Spark across cloud platforms such as AWS and Snowflake. Skilled in ELT/ETL development, dimensional modelling, CDC pipelines, and large-scale data processing using Databricks, Airflow, Kafka, and dbt. Adept at implementing data quality frameworks, observability tooling, and performance optimization for production-grade pipelines. Experienced in deploying automated, infrastructure-as-code workflows using Terraform, CI/CD, and containerized environments. Proven ability to enable analytics, scalability, and real-time data access for enterprise data ecosystems.

SKILLS

- **Programming Languages:** Python, SQL, PySpark, Spark; Bash/Shell
- **Data Engineering & Processing Techniques:** Batch & Streaming Pipelines, Data Ingestion, ELT/ETL Development, Data Modelling (Star & Snowflake), CDC Pipelines, Performance Optimization
- **Big Data & Orchestration Frameworks:** Apache Spark, Databricks, Apache Airflow (DAG Scheduling), Kafka (Event & Real-Time Streaming), dbt (Analytics Engineering)
- **Cloud & Data Platforms:** AWS (S3, Glue, Lambda, EMR), Snowflake, Redshift, BigQuery
- **Databases & Storage:** PostgreSQL, SQL Server, DynamoDB, NoSQL Systems, Delta Lake / Apache Iceberg
- **Data Quality, Governance & Observability:** dbt Tests, Data Validation, Schema Evolution, Great Expectations, Logging & Monitoring
- **DevOps & Pipeline Automation:** Git, Docker, CI/CD Workflows, Terraform (Data Infrastructure), Event-Driven Systems (Kinesis, EventBridge)
- **Business & Collaboration Skills:** Cross-Functional Communication, Requirements Translation, Agile/Scrum Collaboration

WORK EXPERIENCE

DXC Technology

Jan 2025 – Present

Data Engineer

- Built scalable batch and streaming pipelines using Spark and PySpark to process enterprise operational data, enabling reliable ingestion into AWS S3-based data lakes.
- Developed ELT workflows in dbt and Snowflake to transform raw datasets into analytics-ready fact and dimension models aligned with business requirements.
- Orchestrated end-to-end data pipelines with Apache Airflow, implementing modular DAGs, dependency management, and alerting for improved pipeline stability.
- Designed and implemented CDC pipelines leveraging AWS Glue and Kafka to support near-real-time updates across downstream reporting systems.
- Improved pipeline performance by optimizing Spark configurations, partitioning strategies, and file formats, reducing overall processing time by 28%.
- Applied data quality frameworks such as Great Expectations and dbt tests to enforce validation rules, schema checks, and observability coverage.
- Automated deployment and infrastructure provisioning using Terraform and CI/CD workflows, increasing deployment consistency across environments by 35%.

VMware

Jan 2022 – July 2023

Data Engineer

- Engineered distributed data processing pipelines using PySpark and Databricks to integrate product telemetry, customer usage, and platform logs into a unified data lake.
- Designed star and snowflake schema models in Snowflake to support analytics workflows for product, finance, and customer success teams.
- Built automated ingestion frameworks leveraging Kafka and AWS Glue to streamline ingestion of high-volume event and batch datasets.
- Developed reusable dbt models and transformation layers to standardize business logic and improve data consistency across reporting domains.
- Implemented Airflow DAGs with SLA monitoring, dependency controls, and retry logic, increasing pipeline reliability by 22%.
- Strengthened data quality by integrating validation checks, schema enforcement, and anomaly detection rules using Great Expectations.
- Improved pipeline cost efficiency by optimizing compute clusters, storage formats, and partition strategies, reducing overall resource usage by 18%.

Tatvasoft
Data Engineer

Oct 2020 – Jan 2022

- Developed end-to-end ETL pipelines using Python, SQL, and Spark to integrate transactional, CRM, and third-party datasets into centralized data platforms.
- Designed and maintained relational and dimensional data models across PostgreSQL and SQL Server to support analytics and reporting needs.
- Built ingestion workflows using AWS Lambda, Glue, and S3 to automate data acquisition from APIs, flat files, and cloud sources.
- Implemented transformation layers with dbt to standardize business rules, improve data consistency, and simplify downstream consumption.
- Enhanced pipeline performance through query tuning, optimized Spark jobs, and partitioning strategies, improving runtime by 24%.
- Strengthened data reliability by introducing validation checks, schema enforcement, and automated alerts for upstream data anomalies.
- Supported Agile delivery by collaborating with cross-functional teams, ensuring requirements were translated into scalable data solutions that reduced manual reporting work by 30%.

Trigent Software
Data Engineer

July 2016 – Oct 2020

- Built ETL workflows using Python, SQL, and SSIS to consolidate operational, finance, and customer datasets into structured data repositories.
- Designed normalized and dimensional data models in SQL Server to support analytical reporting and self-service BI dashboards.
- Integrated external data sources through REST API ingestion pipelines, automating data collection and reducing manual dependency by 35%.
- Developed scalable processing jobs using PySpark to transform large datasets, apply business rules, and prepare outputs for downstream consumers.
- Implemented automated data validation routines, schema checks, and lineage tracking to improve data accuracy across multiple domains.
- Migrated legacy pipelines to AWS S3 and Glue, enhancing scalability and reducing infrastructure overhead by 20%.
- Collaborated with product, QA, and analytics teams to refine requirements and deliver reliable datasets aligned with evolving business needs.

PROJECTS

Real-Time Event Streaming Pipeline for Customer Activity Analytics

- Built a real-time ingestion pipeline using Kafka, Spark Structured Streaming, and AWS Kinesis to capture high-volume customer activity events from multiple digital channels.
- Designed a Delta Lake-based storage layer on Databricks with optimized partitions, schema enforcement, and MERGE operations for near-real-time analytics.
- Developed downstream enrichment and aggregation jobs using PySpark and dbt, enabling product teams to access actionable metrics through Snowflake dashboards.

Automated Data Quality & Validation Framework for Enterprise Pipelines

- Engineered a modular data quality framework using Python, Great Expectations, and dbt tests to validate schemas, completeness, and business rules across ETL/ELT pipelines.
- Integrated validation checkpoints into Airflow DAGs with automated alerts, logging, and failure handling to improve reliability of AWS S3 and Redshift ingestion workflows.
- Implemented historical anomaly detection using SQL and metadata tracking, reducing recurring data issues and enhancing observability across multiple data domains.

EDUCATION

Master in Computer Information Science

Aug 2023 – Dec 2024

Roosevelt University, Chicago, IL | **GPA: 3.6**